Problem Set 3 Solution

Part One: Computer exercises

1. Draw `n` values from a uniform distribution between `-2` and `2` and calculate the sample mean. Do this `200` times for `n` = `2`, `10`, `50`, and `100`. Make a histogram for each case. What do you find as `n` gets larger?

```
set.seed(123)

simulation_size <- 200
draw_size <- c(2, 10, 50, 100)

df <- data.frame()  # create a dataframe to store the simulation outcomes

for (n in draw_size){
  sample_mean <- replicate(200, mean(runif(n, min = -2, max = 2)))

  # store the outcome in a temporary dataframe
  # temp should be a 200 x 2 df
  temp <- data.frame(
    draw_size = n,
    sample_mean = sample_mean
  )

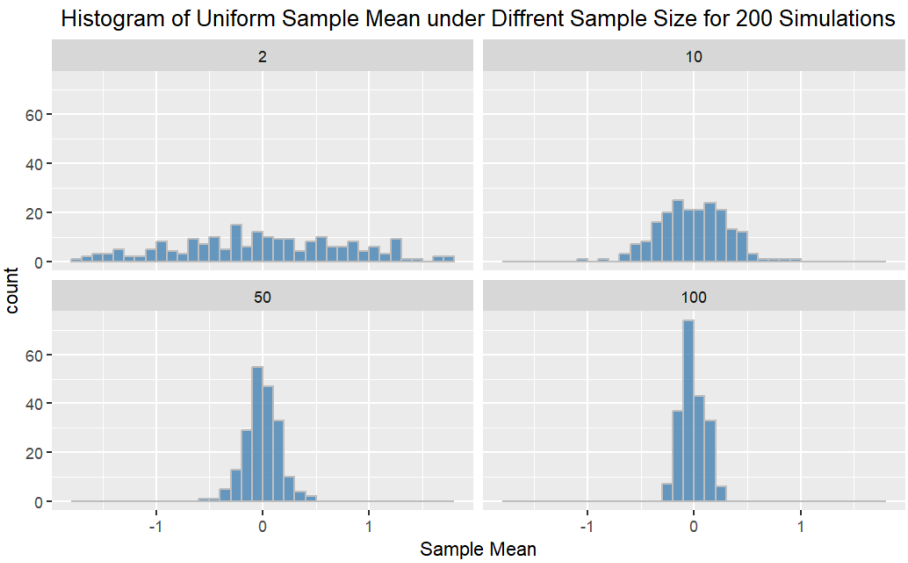
  # combine the temp simulation results with our df
  # after the for loop completes, df will be a 800 x 2 df
  df <- rbind(df, temp)
}

# draw histograms

library(ggplot2)

ggplot(df, aes(x = sample_mean)) +
  geom_histogram(binwidth = 0.1,
                 fill = "steelblue",
                 color = "gray",
                 alpha = 0.8,
                 boundary = 0.1) +
  facet_wrap(~draw_size) +     # facet_wrap() create subplots within a fig
ure
  labs(
```

```
    x = "Sample Mean",
    y = "count",
    title = "Histogram of Uniform Sample Mean under Diffrent Sample Size f
or 200 Simulations"
  ) +
  theme(plot.title = element_text(hjust = 0.5))
```



2. Draw `n` values from an exponential distribution with `λ` = `1.5` and calculate the sample mean. Do this `200` times for `n` = `2`, `10`, `50`, and `100`. Make a histogram of the sample means for each case. What do you find as `n` gets larger?

```
# the code is very similar to Quesiton 1!
# you only need to adjust the sampling function and the parameter

set.seed(123)

simulation_size <- 200
draw_size <- c(2, 10, 50, 100)

lambda <- 1.5

df <- data.frame()  # create a dataframe to store the simulation outcomes

for (n in draw_size){
  sample_mean <- replicate(200, mean(rexp(n, rate = lambda)))

  # store the outcome in a temporary dataframe
  # temp should be a 200 x 2 df
```

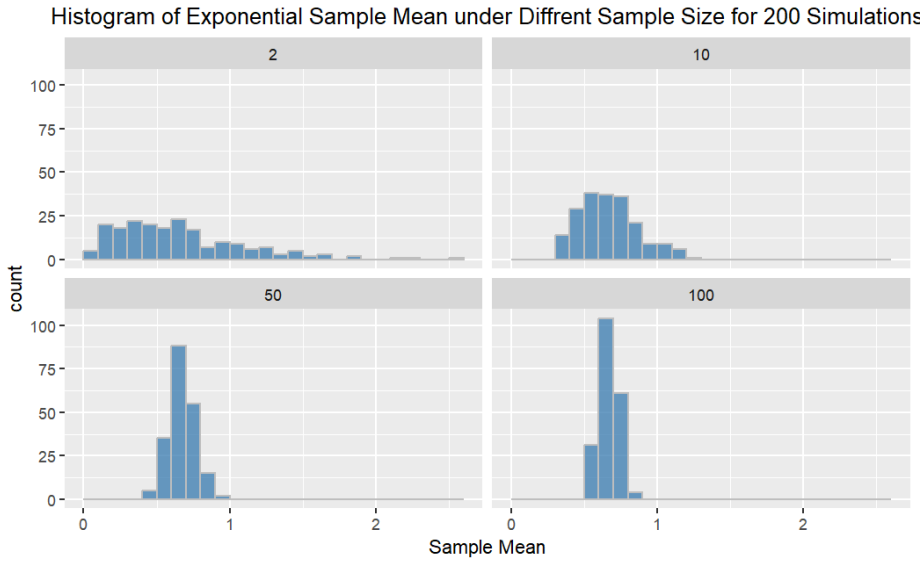
```
  temp <- data.frame(
    draw_size = n,
    sample_mean = sample_mean
  )

  # combine the temp simulation results with our df
  # after the for loop completes, df will be a 800 x 2 df
  df <- rbind(df, temp)
}

# draw histograms

library(ggplot2)

ggplot(df, aes(x = sample_mean)) +
  geom_histogram(binwidth = 0.1,
                 fill = "steelblue",
                 color = "gray",
                 alpha = 0.8,
                 boundary = 0.1) +
  facet_wrap(~draw_size) +     # facet_wrap() create subplots within a fig
ure
  labs(
    x = "Sample Mean",
    y = "count",
    title = "Histogram of Exponential Sample Mean under Diffrent Sample Si
ze for 200 Simulations"
  ) +
  theme(plot.title = element_text(hjust = 0.5))
```



Part Two: Textbook exercises

Question 5.8

Recall that the general formula is **point estimate** $\pm z^* \times SE$.

First, identify the three values. The point estimate is **52%**, $z^* = 2.58$for a 99% confidence level, and $SE = 2.4$

$$52\% \pm 2.58 \times 2.4\% \ \rightarrow \ (45.8\%, 58.2\%)$$

We are 99% confident that **45.8% to 58.2%** of U.S. adult Twitter users get some news on Twitter.

Question 5.10

a. **False.** 50% is included in the 99% confidence interval, hence a null hypothesis of would **not** be rejected at this level.

b. **False.** The standard error measures the variability of the sample proportion and is unrelated to the proportion of the population included in the study.

c. **False.** We need to **increase** the sample size to **decrease** the standard error.

d. **False.** As the confidence level decreases, so does the margin of error, and hence the width of the confidence interval.

Question 5.18

First, the hypotheses should be about the **population proportion** (p), not the sample proportion ($\hat{p}$).

Second, the null value should be what we are testing (25%), not the observed value (24%).

The correct way to set up these hypotheses is

$$\begin{cases} H_0 : p = 0.25 \\ H_A : p \neq 0.25 \end{cases}$$

Question 5.26

a. **Scenario I is higher.** A sample mean based on **less** data tends to be less accurate and has a larger standard error.

b. **Scenario I is higher.** A higher confidence level a larger margin of error.

c. **They are equal.** For a given z-score the p-value depends only on the value, not on the sample size.

d. **Scenario I is higher.** If the null is harder to reject (smaller α), we are more likely to make a Type II error when the alternative is true.

Question 5.32

Prepare — Hypotheses

$$\begin{cases} H_0 : p = 0.08 \\ H_A : p \neq 0.08 \end{cases}$$

Use significance level $\alpha = 0.05$.

Check

Simple random sample $\Rightarrow$ independence.

Success–failure condition: $21 \geq 10$ and $194 - 21 = 173 \geq 10$.

Calculate

$$\hat{p} = \frac{21}{194} = 0.108$$
$$SE = \sqrt{\frac{0.08(1-0.08)}{194}} = 0.0195$$
$$z = \frac{0.108 - 0.08}{0.0195} = 1.44$$

A one–tail area is about 0.075, so the two–sided p-value is $2 \times 0.075 \approx 0.15$.

Conclude

Because the p-value (0.15) is greater than $\alpha = 0.05$, we **do not reject** H_0.

There is not convincing evidence that the fraction of children who are nearsighted differs from 0.08.

Question 5.36

As the sample size increases, the **standard error decreases**. For the same true effect, the test statistic (e.g., z) **increases**, and the **p-value decreases**.